

K.MANICKAPERUMAL

MERN Stack Engineer

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PROFESSIONAL SUMMARY

Performance-driven MERN Stack Engineer with **3+** years of experience architecting scalable backend systems and optimizing data engines. Leveraging deep expertise in Node.js, TypeScript, and advanced algorithmic problem-solving to build high-performance, responsive user interfaces in React.js. Proven ability to own features end-to-end, from efficient database schema design to seamless frontend state management.

TECHNICAL SKILLS

Language - Typescript, Javascript, Python

Frontend - React.js, Redux/Context API, Next JS

Backend & Frameworks- Node js, Nest js, Express js, Jest

Databases & ORM - MongoDB, PostgreSQL, MySQL, TypeORM, Knex js

Infrastructure & Tools - AWS, Docker, Redis, CI/CD, RESTful APIs, Git, Linux

PROFESSIONAL EXPERIENCE

Lumel Technologies Pvt Ltd

Associate Product Developer

Chennai

June 2025 - Feb 2026

- **Performance Engineering** - Re-engineered a mission critical data deletion engine by replacing nested application level loops with an optimized SQL stored procedure, **achieving a 99% reduction** in time complexity from $O(n^3)$ to $O(n)$ and eliminating database timeout errors during peak loads.
- **Scalable Architecture** - Designed and decoupled a **dynamic filtering module** capable of processing multi million row datasets with sub-second latency.

Megsoft Business Solution Pvt Ltd

Software Engineer

Madurai

Sep 2023 - Dec 2024

- **Infrastructure Cost Optimization** - Eliminated **100% of AWS messaging costs** by architecting and deploying a custom **Internal queue management system**.

Altrosyn Technologies Pvt Ltd

Junior Software Engineer

Chennai

May 2022 - July 2023

- **Distributed Processing** - Engineered a high-throughput **Bulk Data Ingestion Engine** capable of processing **5+ million rows** by implementing a multi-process architecture; ensured system stability and 0% memory overflow during massive data loads.

- **DevOps & Reliability:** Managed the end-to-end deployment of Node.js executable binaries on **Linux production environments**; performed root-cause analysis (RCA) and resolved critical production bugs to maintain **99.9% application uptime**.
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PROJECTS

Distributed Code Execution Engine (Leetcode clone)

[Github Link](#)

- **Architected a microservices-based submission system** using Node.js and TypeScript, decoupling problem management from code execution to ensure system scalability and fault tolerance.
 - **Implemented asynchronous task processing** by integrating **Redis** with **BullMQ**, enabling the submission service to offload heavy code evaluation tasks to a background worker service
 - **Enforced resource constraints and security** by configuring Docker host limits, including **CPU quotas**, **memory limits**, and **PID limits** to prevent malicious code (e.g., fork bombs) from exhausting host resources.
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ACHIEVEMENTS & CERTIFICATIONS

React - The Complete Guide | Udemy

[Certificate Link](#)

- Advanced coursework covering modern React features, component lifecycle hooks, state management patterns, Redux and front-end optimization strategies.

Data Structures & Algorithms | Udemy

[Certificate Link](#)

- Advanced coursework covering algorithmic analysis ($O(N)$ time and space complexity), dynamic programming, and the implementation of complex data structures (Trees, Graphs, Hash Tables).

Advanced Problem Solving | Leetcode, Hacker Rank

[Certificate Link](#)

- Solved **250+** highly curated problems on LeetCode, demonstrating strong data structures and algorithms (DSA) proficiency in optimization and runtime complexity $O(N)$
 - Earned the HackerRank Problem Solving Certificate, validating advanced algorithmic execution and problem-solving capabilities under production-level constraints.
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EDUCATION

Kamaraj College, Tuticorn
Bachelors of computer science

Manonmaniam sundaranar University
CGPA - 7.50